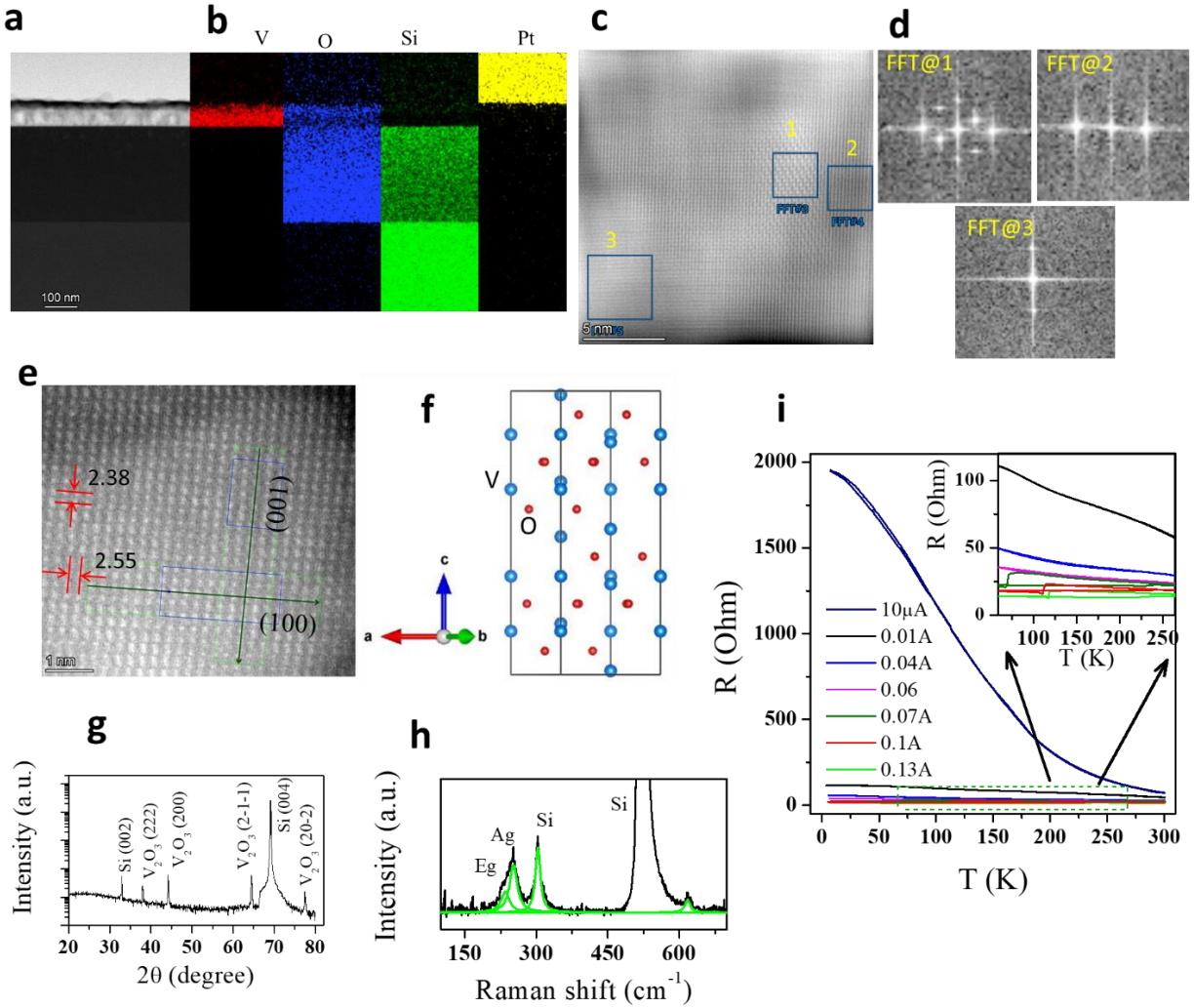

Fig. 1| Characterization and electrical response of the V_2O_3/Si thin film. **a** & **b,** Cross sectional STEM HAADF image and EDS mapping. **c,** Large area STEM image showing disordered regions. **d,** Fast Fourier Transformation of the selected area of Fig. **c**. **e,** Atomically resolved STEM image on *a-c* plane. **f,** V_2O_3 schematic. **g** & **h,** XRD and Raman spectra. **i,** Resistance R vs T under different dc currents from 10μA to 0.13A applied at 300K and resistance was measured in both heating and cooling cycles with temperature sweeping rate 3K/min. Inset shows the magnified view of the rectangular green dotted area.

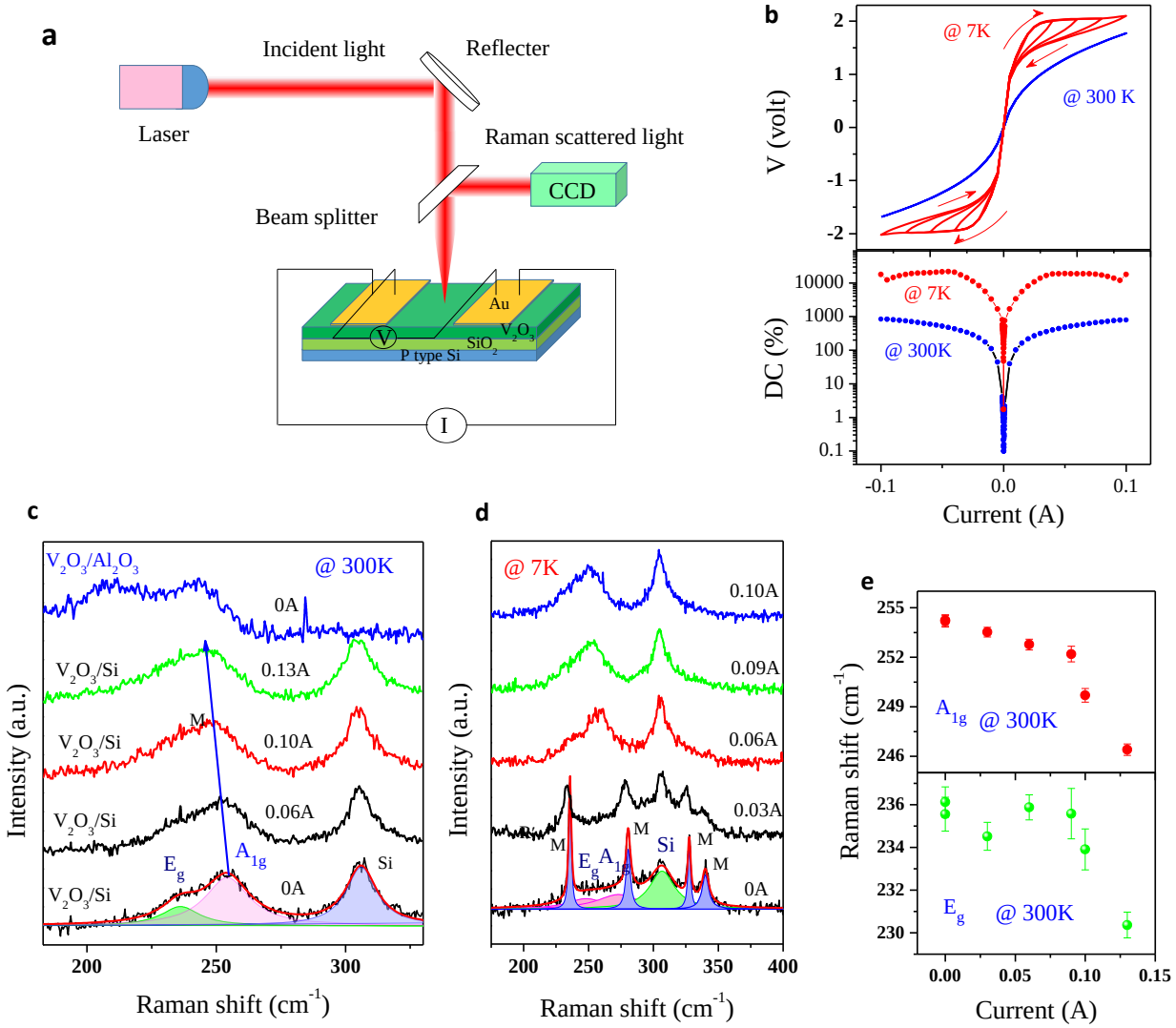

Fig. 2| *In-situ* electric field dependent Raman and V-I characteristics of V_2O_3/Si thin film at 300K and 7K. a, Schematic of the experimental geometry for simultaneous Raman and resistance measurements. **b,** V-I characteristic along with minor hysteresis loops and differential conductance (DC) as a function of applied electric current. **c,** Raman spectra at 300K, measured on V_2O_3/Si under different electric fields (currents) along with Raman spectrum of metallic V_2O_3/Al_2O_3 (top curve) shown for comparison. **d**, Raman spectra at 7K under field. Symbol M depicts the modes corresponding to the monoclinic phase ($C_{2/c}$). **e,** Variation of A_{1g} and E_g mode position (rhombohedral R′ phase) at 300K as a function of applied electric current.

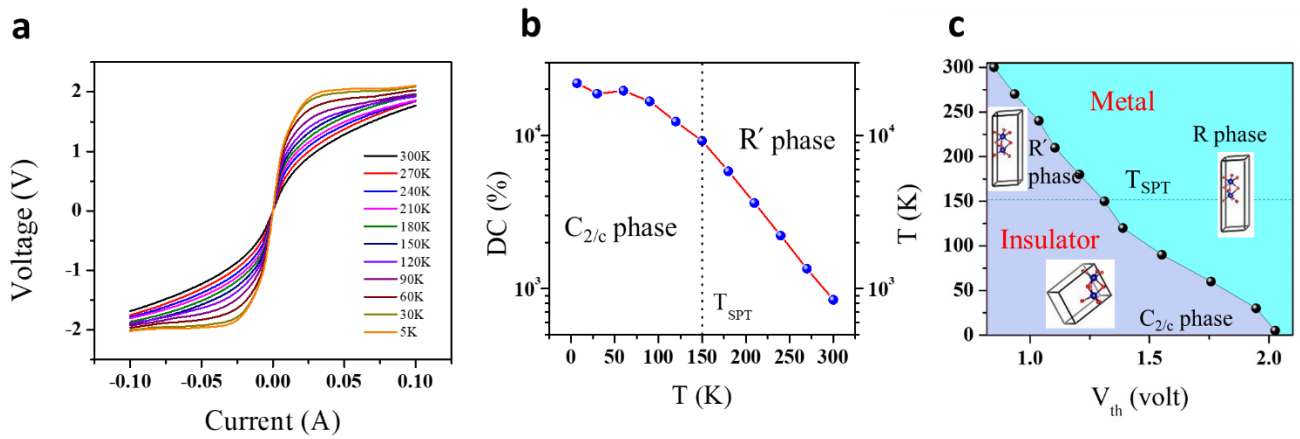

Fig. 3| Electric field induced Differential Conductance (DC) change at different temperatures and IMT phase diagram of V_2O_3/Si thin film: a, Temperature evolution of I-V characteristics of V_2O_3/Si thin film. **b,** Differential conductance change due to application of 100mA current at different temperatures. **c,** Phase diagram of the electric field induced IMT as a function of temperature. Threshold voltage V_{th} of IMT decreases with increasing temperature. Separation between two electrodes "d" was 550μm. Blue dash line represent the onset temperature of structural phase transition (T_{SPT}). Crystal unit cells associated with the insulating and metallic phases are also shown. Both metallic R and insulator R′ phase belong to R-3c structure, but *c/a* ratio is higher in R phase. Monoclinic $C_{2/c}$ phase appears below T_{SPT}.